EXTRACTION OF METALS & CORROSION

Class 10 Science • Chapter 3: Metals and Non-metals • Detailed Notes with Board Questions

1. Occurrence of Metals — Key Terms

The earth's crust is the major source of metals. Sea water also contains soluble salts such as sodium chloride and magnesium chloride.

Term	Meaning	Example
Mineral	Naturally occurring element or compound found in the earth's crust.	Bauxite, haematite
Ore	A mineral from which a metal can be extracted **profitably**. All ores are minerals, but all minerals are not ores.	Cinnabar (HgS), zinc blende (ZnS)
Gangue	Unwanted impurities (soil, sand, rocky matter) present in an ore.	Sand (SiO_2) in ore
Metallurgy	The complete process of extracting a metal from its ore and refining it for use.	—
Native / free state	Metals so unreactive that they are found uncombined in nature.	Gold, silver, platinum, copper (partly)

Types of ores

Type of ore	Metals usually found this way	Examples
Oxide ores	Al, Fe, Sn, Mn, Zn, Cu	Bauxite, haematite, cassiterite, pyrolusite
Sulphide ores	Zn, Pb, Cu, Hg, Ag, Fe	Zinc blende, galena, copper glance, cinnabar
Carbonate ores	Ca, Mg, Zn, Fe, Cu	Limestone, magnesite, calamine, siderite, malachite
Halide (chloride/fluoride) ores	Na, K, Mg, Ca, Ag, Al	Rock salt, sylvite, carnallite, fluorspar, cryolite
Sulphate ores	Ca, Mg, Ba	Gypsum, Epsom salt, barytes
Nitrate ores	Na, K	Chile saltpetre, saltpetre (nitre)

Complete list of important ores

Ores marked ★ are the ones asked most often in board exams.

Metal	Ore (common name)	Formula	Type
Potassium (K)	Sylvite	KCl	Chloride
	Carnallite	$KCl·MgCl_2·6H_2O$	Chloride
	Saltpetre (nitre)	KNO_3	Nitrate
Sodium (Na)	★ Rock salt	NaCl	Chloride

Metal	Ore (common name)	Formula	Type
	Chile saltpetre	$NaNO_3$	Nitrate
	Borax	$Na_2B_4O_7 \cdot 10H_2O$	Borate
Calcium (Ca)	★ Limestone	$CaCO_3$	Carbonate
	Gypsum	$CaSO_4 \cdot 2H_2O$	Sulphate
	Fluorspar	CaF_2	Fluoride
	Dolomite (Ca and Mg)	$CaCO_3 \cdot MgCO_3$	Carbonate
Magnesium (Mg)	Magnesite	$MgCO_3$	Carbonate
	Epsom salt	$MgSO_4 \cdot 7H_2O$	Sulphate
	Carnallite	$KCl \cdot MgCl_2 \cdot 6H_2O$	Chloride
Aluminium (Al)	★ Bauxite	$Al_2O_3 \cdot 2H_2O$	Oxide
	Cryolite	Na_3AlF_6	Fluoride
	Corundum	Al_2O_3	Oxide
Zinc (Zn)	★ Zinc blende	ZnS	Sulphide
	★ Calamine	$ZnCO_3$	Carbonate
	Zincite	ZnO	Oxide
Iron (Fe)	★ Haematite	Fe_2O_3	Oxide
	Magnetite	Fe_3O_4	Oxide
	Limonite	$2Fe_2O_3 \cdot 3H_2O$	Oxide
	Siderite	$FeCO_3$	Carbonate
	Iron pyrites (not used for iron)	FeS_2	Sulphide
Tin (Sn)	Cassiterite (tinstone)	SnO_2	Oxide
Lead (Pb)	★ Galena	PbS	Sulphide
Copper (Cu)	★ Copper glance	Cu_2S	Sulphide
	Copper pyrites	$CuFeS_2$	Sulphide
	Cuprite (ruby copper)	Cu_2O	Oxide
	Malachite	$CuCO_3 \cdot Cu(OH)_2$	Carbonate
Mercury (Hg)	★ Cinnabar	HgS	Sulphide
Silver (Ag)	Argentite (silver glance)	Ag_2S	Sulphide
	Horn silver	$AgCl$	Chloride
Manganese (Mn)	Pyrolusite	MnO_2	Oxide
Chromium (Cr)	Chromite	$FeO \cdot Cr_2O_3$	Oxide
Gold, Platinum	Mostly native (free)	Au, Pt	—

Remember: Metals at the **bottom** of the activity series (Au, Ag, Pt) are least reactive → found **free**. Metals at the **top** (K, Na, Ca, Mg, Al) are so reactive they are **never** found free. Many ores are **oxides**,

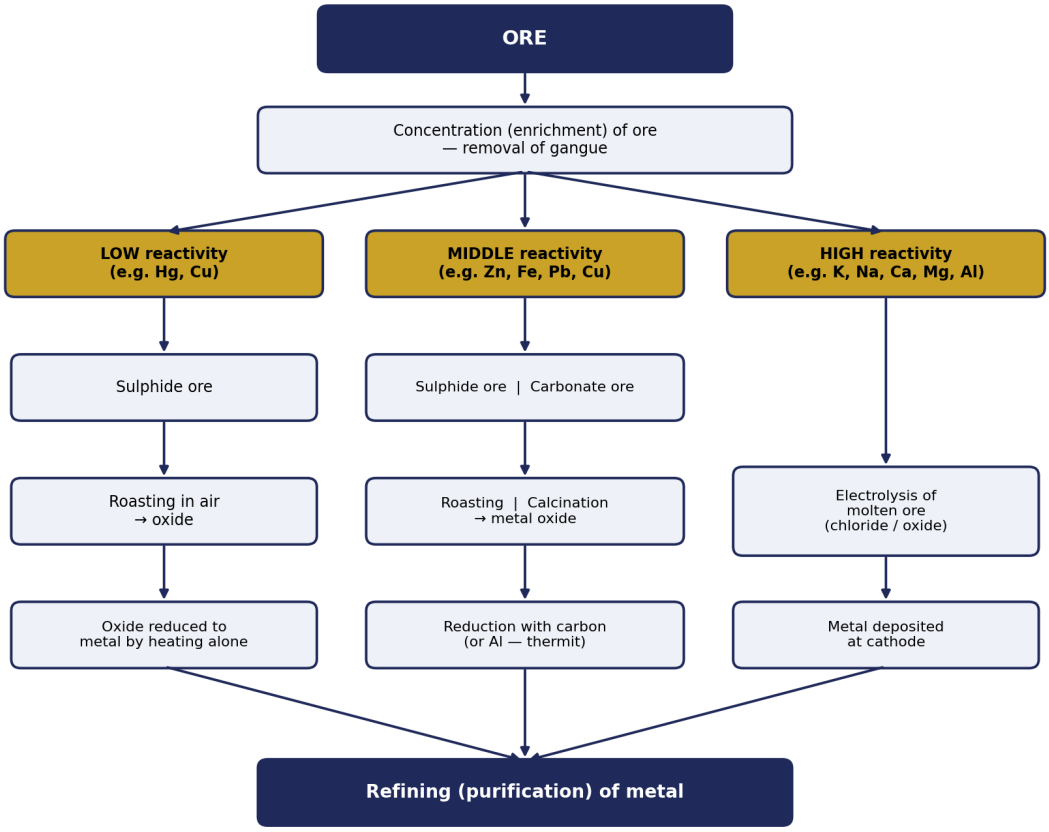
> because oxygen is abundant and reactive; sulphide and carbonate ores are converted to oxides before reduction.

2. Activity Series and Method of Extraction

The method used to extract a metal depends on its **position in the activity series** (its reactivity).

Reactivity	Metals	Method of extraction
High	K, Na, Ca, Mg, Al	Electrolytic reduction (electrolysis of molten ore)
Middle	Zn, Fe, Pb, Cu	Roasting / calcination → oxide, then reduction with carbon (coke) or a more reactive metal
Low	Hg, Cu (from Cu_2S)	Heating the sulphide ore in air (self-reduction)
Very low	Ag, Au, Pt	Found in free (native) state; only need purification

Steps involved in the extraction of metals (flowchart)



Every extraction involves three broad steps:

1. **Enrichment (concentration) of the ore** — removing gangue.
2. **Extraction of the metal** — converting the concentrated ore to metal (method depends on reactivity).
3. **Refining** — purifying the impure metal.

3. Enrichment (Concentration) of Ores

Ores mined from the earth contain large amounts of gangue. It must be removed **before** extraction. The method depends on the difference in physical or chemical properties between the gangue and the ore.

Method	Principle	Used for
Hydraulic washing (gravity separation)	Heavier ore particles settle; lighter gangue is washed away by water.	Oxide ores (e.g. haematite)
Froth flotation	Ore particles are wetted by oil and rise with the froth; gangue is wetted by water and sinks.	Sulphide ores (e.g. ZnS, Cu_2S)
Magnetic separation	Magnetic ore is attracted by a magnet; non-magnetic gangue is not.	Magnetic ores (e.g. magnetite)

4. Extracting Metals Low in the Activity Series

Oxides of these metals are unstable; they can be reduced to the metal **by heating alone**.

(a) Mercury from cinnabar (HgS)

Cinnabar is heated in air. It is first converted to mercuric oxide, which on further heating is reduced to mercury.

> $2HgS(s) + 3O_2(g) \rightarrow 2HgO(s) + 2SO_2(g)$ [Heat]
>
> $2HgO(s) \rightarrow 2Hg(l) + O_2(g)$ [Heat]

(b) Copper from copper glance (Cu_2S)

Copper glance is heated in air. Part of it is oxidised to copper(I) oxide, which then reacts with the remaining Cu_2S to give copper metal (**self-reduction**).

> $2Cu_2S(s) + 3O_2(g) \rightarrow 2Cu_2O(s) + 2SO_2(g)$ [Heat]
>
> $2Cu_2O(s) + Cu_2S(s) \rightarrow 6Cu(s) + SO_2(g)$ [Heat]

> **Board favourite:** "How is mercury obtained from its ore? Write the equations." — Always write **both** steps with balanced equations and mention that heating alone is enough because mercury is low in the activity series.

5. Extracting Metals in the Middle of the Activity Series

Metals like Zn, Fe, Pb and Cu are moderately reactive. They are usually found as **sulphides or carbonates**. It is easier to obtain a metal from its **oxide** than from its sulphide or carbonate, so the ore is first converted to an oxide.

Step 1: Conversion to oxide — Roasting and Calcination

Point of difference	Roasting	Calcination
Type of ore	**Sulphide** ore	**Carbonate** ore
Condition	Heated strongly in the presence of **excess air**	Heated strongly in **limited air** (or absence of air)
Gas given out	Sulphur dioxide (SO_2)	Carbon dioxide (CO_2)
Example	$2ZnS + 3O_2 \rightarrow 2ZnO + 2SO_2$	$ZnCO_3 \rightarrow ZnO + CO_2$

Equations for zinc (asked very often):

> Roasting: $2ZnS(s) + 3O_2(g) \rightarrow 2ZnO(s) + 2SO_2(g)$ [Heat]
>
> Calcination: $ZnCO_3(s) \rightarrow ZnO(s) + CO_2(g)$ [Heat]

More examples of roasting and calcination:

Ore	Process	Equation
Galena (PbS)	Roasting	$2PbS + 3O_2 \rightarrow 2PbO + 2SO_2$
Cinnabar (HgS)	Roasting	$2HgS + 3O_2 \rightarrow 2HgO + 2SO_2$
Copper glance (Cu_2S)	Roasting	$2Cu_2S + 3O_2 \rightarrow 2Cu_2O + 2SO_2$
Siderite ($FeCO_3$)	Calcination	$FeCO_3 \rightarrow FeO + CO_2$
Malachite	Calcination	$CuCO_3{\cdot}Cu(OH)_2 \rightarrow 2CuO + CO_2 + H_2O$
Limestone ($CaCO_3$)	Calcination	$CaCO_3 \rightarrow CaO + CO_2$
Magnesite ($MgCO_3$)	Calcination	$MgCO_3 \rightarrow MgO + CO_2$
Bauxite ($Al_2O_3{\cdot}2H_2O$)	Calcination (removes water)	$Al_2O_3{\cdot}2H_2O \rightarrow Al_2O_3 + 2H_2O$

> **Tip:** Calcination also removes **water of crystallisation** (as in bauxite) and volatile impurities, not just CO_2. CaO and MgO from calcination are **not** reduced by carbon — Ca and Mg are obtained by electrolysis.

Step 2: Reduction of the oxide

(i) Reduction with carbon (coke): Carbon acts as the reducing agent and removes oxygen from the metal oxide.

> $ZnO(s) + C(s) \rightarrow Zn(s) + CO(g)$

(ii) Reduction with a more reactive metal (displacement): Highly reactive metals such as sodium, calcium and aluminium can displace less reactive metals from their compounds. These reactions are **highly exothermic** — so much heat is released that the metal is produced in the **molten state**.

> $3MnO_2(s) + 4Al(s) \rightarrow 3Mn(l) + 2Al_2O_3(s) + Heat$

Thermit reaction

The reaction of **iron(III) oxide with aluminium** is called the thermit reaction. It produces molten iron and is used to **join railway tracks** and **cracked machine parts** (thermit welding).

> $Fe_2O_3(s) + 2Al(s) \rightarrow 2Fe(l) + Al_2O_3(s) + Heat$

> **Why aluminium?** Aluminium is **more reactive** than iron and has a greater affinity for oxygen, so it removes oxygen from Fe_2O_3. Here Fe_2O_3 is **reduced** and Al is **oxidised**; aluminium is the reducing agent.

Other displacement reactions using aluminium:

> $Cr_2O_3(s) + 2Al(s) \rightarrow 2Cr(l) + Al_2O_3(s) + Heat$
>
> $3Fe_3O_4(s) + 8Al(s) \rightarrow 9Fe(l) + 4Al_2O_3(s) + Heat$

Solved examples — metals of middle reactivity

Example 1: Zinc from zincite (ZnO) — the ore is already an oxide, so only reduction is needed.

> $ZnO(s) + C(s) \rightarrow Zn(s) + CO(g)$

Example 2: Lead from galena (PbS) — sulphide ore, so roasting first, then reduction with carbon.

> Roasting: $2PbS(s) + 3O_2(g) \rightarrow 2PbO(s) + 2SO_2(g)$

> Reduction: $PbO(s) + C(s) \rightarrow Pb(s) + CO(g)$

Example 3: Iron from haematite (Fe_2O_3) — oxide ore, reduced with coke (carbon). In industry this is done in a **blast furnace**, where coke first forms carbon monoxide, which reduces the oxide.

> $Fe_2O_3(s) + 3C(s) \rightarrow 2Fe(l) + 3CO(g)$
>
> $C + O_2 \rightarrow CO_2$; $CO_2 + C \rightarrow 2CO$; $Fe_2O_3 + 3CO \rightarrow 2Fe + 3CO_2$

Limestone is added as a **flux** to remove sandy gangue as **slag**: $CaCO_3 \rightarrow CaO + CO_2$; $CaO + SiO_2 \rightarrow CaSiO_3$ (slag).

Example 4: Iron from siderite ($FeCO_3$) — carbonate ore, so calcination first.

> Calcination: $FeCO_3(s) \rightarrow FeO(s) + CO_2(g)$
>
> Reduction: $FeO(s) + C(s) \rightarrow Fe(s) + CO(g)$

Example 5: Copper from malachite — carbonate ore, calcined to copper(II) oxide, then reduced.

> Calcination: $CuCO_3{\cdot}Cu(OH)_2(s) \rightarrow 2CuO(s) + CO_2(g) + H_2O(g)$
>
> Reduction: $CuO(s) + C(s) \rightarrow Cu(s) + CO(g)$

Example 6: Tin from cassiterite (SnO_2) — oxide ore, reduced directly with carbon.

> $SnO_2(s) + 2C(s) \rightarrow Sn(s) + 2CO(g)$

Example 7: Manganese from pyrolusite (MnO_2) — reduced by aluminium (see displacement above).

> **How to decide the steps:** Look at the ore type. **Sulphide → roasting**, **carbonate → calcination**, **oxide → go straight to reduction**. Then check reactivity: middle metals → carbon or aluminium; high metals → electrolysis.

6. Extracting Metals High in the Activity Series

Metals like Na, K, Ca, Mg and Al **cannot be obtained by heating their oxides with carbon**, because they have **more affinity for oxygen than carbon** does. They are obtained by **electrolytic reduction** of their **molten** chlorides or oxides.

During electrolysis, the metal is deposited at the **cathode** (negative electrode) and a non-metal is liberated at the **anode** (positive electrode).

Electrolysis of molten sodium chloride

> At cathode: $Na^+ + e^- \rightarrow Na$ (reduction)
>
> At anode: $2Cl^- \rightarrow Cl_2 + 2e^-$ (oxidation)

More examples of electrolytic reduction

Potassium — from molten potassium chloride (from sylvite/carnallite):

> Cathode: $K^+ + e^- \rightarrow K$ Anode: $2Cl^- \rightarrow Cl_2 + 2e^-$

Magnesium — from molten magnesium chloride (from carnallite or sea water):

> Cathode: $Mg^{2+} + 2e^- \rightarrow Mg$ Anode: $2Cl^- \rightarrow Cl_2 + 2e^-$

Calcium — from molten calcium chloride:

> Cathode: $Ca^{2+} + 2e^- \rightarrow Ca$ Anode: $2Cl^- \rightarrow Cl_2 + 2e^-$

Aluminium — by electrolytic reduction of **alumina** (Al_2O_3) obtained from bauxite. Alumina is dissolved in molten **cryolite**, which lowers its melting point and increases conductivity. Carbon (graphite) is used as the electrodes.

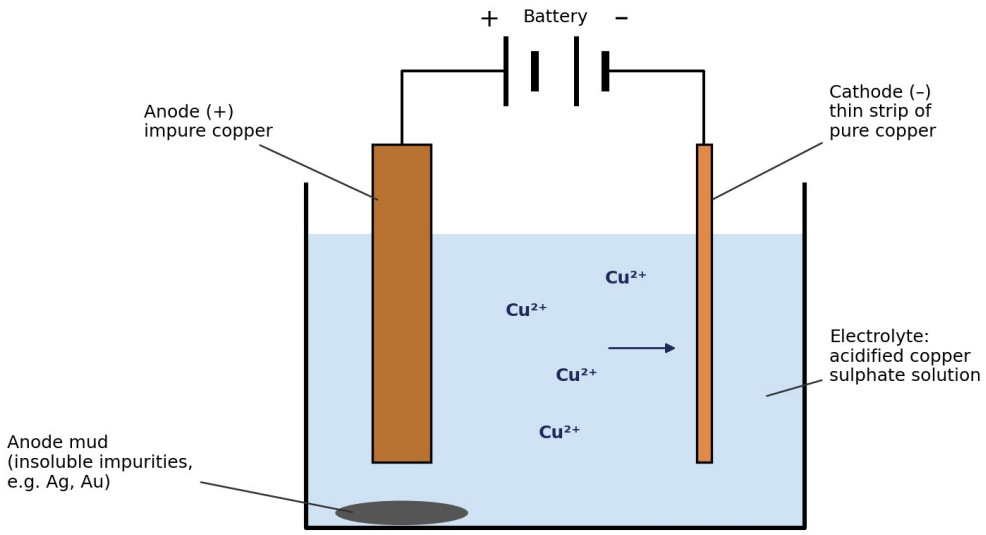
> Cathode: $Al^{3+} + 3e^- \rightarrow Al$ Anode: $2O^{2-} \rightarrow O_2 + 4e^-$

The oxygen released at the anode reacts with the carbon anode ($C + O_2 \rightarrow CO_2$), so the anodes are replaced from time to time.

> **Note:** The ore must be **molten**, not an aqueous solution. In water, H^+ ions are discharged at the cathode instead, and hydrogen is formed rather than sodium.

7. Refining of Metals — Electrolytic Refining

Metals obtained by reduction contain impurities. The most widely used method of purification is **electrolytic refining**. Metals such as **copper, zinc, tin, nickel, silver and gold** are refined this way.



Electrolytic refining of copper

Part of cell	What is used
Anode (+)	Thick block of **impure** metal
Cathode (–)	Thin strip of **pure** metal
Electrolyte	Solution of a salt of the metal — for copper: **acidified copper sulphate** solution

Working

- On passing current, the **impure metal from the anode dissolves** into the electrolyte as ions.
- An **equal amount of pure metal** from the electrolyte is **deposited on the cathode**, which grows thicker.
- **Soluble impurities** go into the solution.
- **Insoluble impurities** settle at the bottom of the anode as **anode mud** (contains valuable metals like silver and gold).

> At anode: $Cu(s) \rightarrow Cu^{2+}(aq) + 2e^-$
>
> At cathode: $Cu^{2+}(aq) + 2e^- \rightarrow Cu(s)$

8. Corrosion

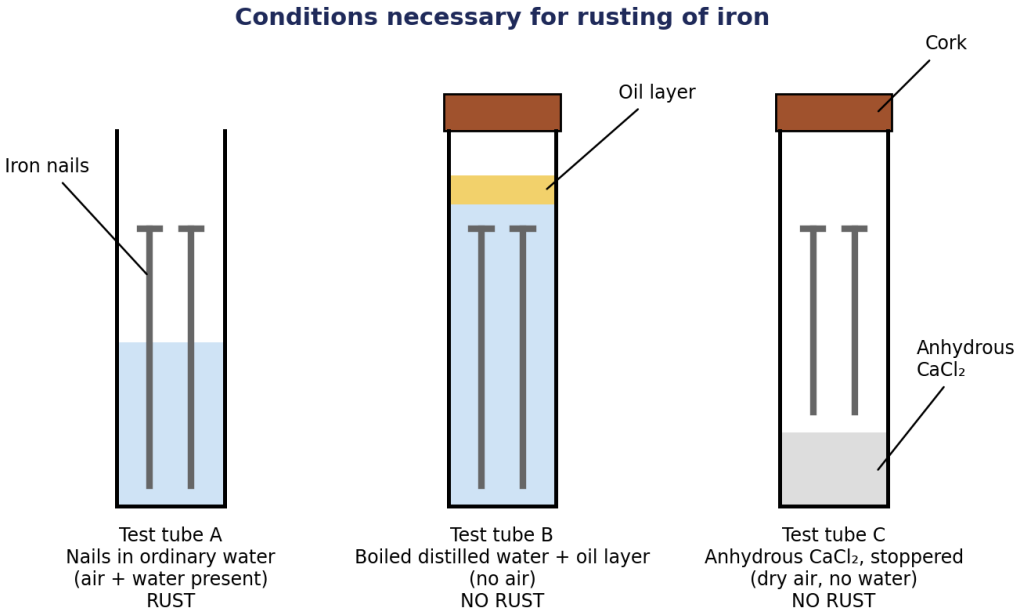
Corrosion is the slow eating away of a metal when its surface is attacked by air, moisture, acids or other substances around it.

Metal	What happens	Substance formed
Silver	Turns **black** after some time in air	Silver sulphide (Ag_2S) — reacts with sulphur/H_2S in air
Copper	Loses its shine; gets a **green** coating	Basic copper carbonate — reacts with moist CO_2 in air
Iron	Gets a **reddish-brown** flaky coating (rust)	Hydrated iron(III) oxide, $Fe_2O_3 \cdot xH_2O$

Rusting of iron

Rusting is the corrosion of iron. When iron is exposed to **moist air** for a long time, it gets coated with a reddish-brown substance called **rust** ($Fe_2O_3 \cdot xH_2O$). **Both air (oxygen) and water (moisture) are necessary** for rusting.

Activity: To show that air and water are both necessary for rusting



Test tube	Set-up	Present	Observation
A	Iron nails in ordinary tap water, left open	Air + water	Nails **rust**
B	Nails in **boiled** distilled water (removes dissolved air) with a layer of **oil** on top (stops air dissolving back); corked	Water only	**No rust**
C	Nails with **anhydrous calcium chloride** (absorbs moisture); corked	Dry air only	**No rust**

Conclusion: Iron rusts only when **both air and water** are present.

> **Why each step?** Water is **boiled** to remove dissolved air; **oil** prevents air from dissolving again; **anhydrous $CaCl_2$** absorbs moisture from the air in tube C.

Prevention of corrosion

- **Painting, oiling or greasing** — forms a layer that keeps air and moisture away.
- **Galvanisation** — coating iron or steel with a thin layer of **zinc**. Galvanised iron is protected **even if the zinc coating is broken**, because zinc is more reactive than iron and gets corroded first.

- **Chrome plating / tin plating (electroplating)** — coating with a less easily corroded metal.
- **Anodising** — making the natural protective oxide layer on **aluminium** thicker by electrolysis.
- **Alloying** — mixing the metal with other elements, e.g. **stainless steel** (iron + nickel + chromium) does not rust.

Alloys

An **alloy** is a **homogeneous mixture** of two or more metals, or a metal and a non-metal. It is prepared by melting the primary metal and dissolving the other elements in it in definite proportions, then cooling.

- Pure iron is soft and stretches easily when hot; mixed with a small amount of carbon (about 0.05%) it becomes **hard and strong** (steel).
- If one of the metals is **mercury**, the alloy is called an **amalgam**.
- The **electrical conductivity** and **melting point** of an alloy are **lower** than those of the pure metals.

Alloy	Constituents	Special property / use
Brass	Copper + Zinc	Utensils, decorative items
Bronze	Copper + Tin	Statues, medals
Solder	Lead + Tin	**Low melting point** → used for welding electrical wires together
Stainless steel	Iron + Nickel + Chromium (+ carbon)	Does not rust — utensils, surgical instruments
Amalgam	Mercury + another metal	Dental fillings

> **Gold jewellery:** Pure gold (**24 carat**) is very soft, so it is not suitable for jewellery. It is alloyed with **silver or copper** to make it hard. **22-carat gold** means 22 parts pure gold and 2 parts copper or silver.
>
> **Wonder of ancient Indian metallurgy:** The **iron pillar** near the Qutub Minar in Delhi was built more than 1600 years ago. It is about 8 m high, weighs about 6000 kg, and has **not rusted** — showing the advanced skill of ancient Indian iron workers.

9. Important Equations at a Glance

Process	Equation
Roasting of zinc blende	$2ZnS + 3O_2 \rightarrow 2ZnO + 2SO_2$
Roasting of galena	$2PbS + 3O_2 \rightarrow 2PbO + 2SO_2$
Reduction of lead oxide	$PbO + C \rightarrow Pb + CO$
Reduction of haematite	$Fe_2O_3 + 3C \rightarrow 2Fe + 3CO$
Calcination of siderite	$FeCO_3 \rightarrow FeO + CO_2$
Calcination of malachite	$CuCO_3{\cdot}Cu(OH)_2 \rightarrow 2CuO + CO_2 + H_2O$
Reduction of copper(II) oxide	$CuO + C \rightarrow Cu + CO$
Reduction of cassiterite	$SnO_2 + 2C \rightarrow Sn + 2CO$
Calcination of calamine	$ZnCO_3 \rightarrow ZnO + CO_2$
Reduction of zinc oxide	$ZnO + C \rightarrow Zn + CO$
Roasting of cinnabar	$2HgS + 3O_2 \rightarrow 2HgO + 2SO_2$

Process	Equation
Reduction of mercuric oxide	$2HgO \rightarrow 2Hg + O_2$
Roasting of copper glance	$2Cu_2S + 3O_2 \rightarrow 2Cu_2O + 2SO_2$
Self-reduction of copper	$2Cu_2O + Cu_2S \rightarrow 6Cu + SO_2$
Thermit reaction	$Fe_2O_3 + 2Al \rightarrow 2Fe + Al_2O_3 + Heat$
Displacement by Al	$3MnO_2 + 4Al \rightarrow 3Mn + 2Al_2O_3 + Heat$
Electrolysis of molten NaCl	Cathode: $Na^+ + e^- \rightarrow Na$ Anode: $2Cl^- \rightarrow Cl_2 + 2e^-$
Displacement by Al (chromium)	$Cr_2O_3 + 2Al \rightarrow 2Cr + Al_2O_3 + Heat$
Electrolysis of molten $MgCl_2$	Cathode: $Mg^{2+} + 2e^- \rightarrow Mg$ Anode: $2Cl^- \rightarrow Cl_2 + 2e^-$
Electrolysis of alumina	Cathode: $Al^{3+} + 3e^- \rightarrow Al$ Anode: $2O^{2-} \rightarrow O_2 + 4e^-$
Electrolytic refining of Cu	Anode: $Cu \rightarrow Cu^{2+} + 2e^-$ Cathode: $Cu^{2+} + 2e^- \rightarrow Cu$

10. Frequently Asked Board Questions (with Answers)

These questions follow the patterns repeatedly asked in CBSE board examinations and sample papers on this topic.

A. Multiple Choice Questions (1 mark)

Q1. Which of the following metals is found in the free state in nature?

(a) Sodium (b) Gold (c) Zinc (d) Iron

Ans. (b) Gold

Q2. Cinnabar is an ore of:

(a) Copper (b) Zinc (c) Mercury (d) Iron

Ans. (c) Mercury

Q3. The process of heating a carbonate ore strongly in limited air is called:

(a) Roasting (b) Calcination (c) Reduction (d) Refining

Ans. (b) Calcination

Q4. In the electrolytic refining of copper, the electrolyte used is:

(a) Dilute HCl (b) Molten NaCl (c) Acidified $CuSO_4$ solution (d) $ZnSO_4$ solution

Ans. (c) Acidified $CuSO_4$ solution

Q5. Which metals are commonly found in the anode mud during refining of copper?

(a) Na and K (b) Ag and Au (c) Zn and Fe (d) Ca and Mg

Ans. (b) Ag and Au

Q6. Galvanisation is a method of protecting iron by coating it with:

(a) Tin (b) Copper (c) Zinc (d) Chromium

Ans. (c) Zinc

Q7. The chemical formula of rust is:

(a) Fe_2O_3 (b) Fe_3O_4 (c) $Fe_2O_3 \cdot xH_2O$ (d) FeO

Ans. (c) $Fe_2O_3 \cdot xH_2O$

Q8. Solder is an alloy of:

(a) Cu and Zn (b) Pb and Sn (c) Cu and Sn (d) Fe and Ni

Ans. (b) Pb and Sn

B. Assertion–Reason (1 mark)

(a) Both A and R are true and R is the correct explanation of A. (b) Both A and R are true but R is not the correct explanation of A. (c) A is true but R is false. (d) A is false but R is true.

Q9. A: Sodium and potassium cannot be obtained by reducing their oxides with carbon. R: These metals have more affinity for oxygen than carbon.

Ans. (a)

Q10. A: Galvanised iron does not rust even if the zinc coating is broken. R: Zinc is more reactive than iron.

Ans. (a)

Q11. A: The thermit reaction is used to join railway tracks. R: In the thermit reaction, aluminium is reduced by iron oxide.

Ans. (c) — A is true; R is false because aluminium is **oxidised** (it is the reducing agent).

Q12. A: Metals like gold and silver are found in the free state. R: They are highly reactive metals.

Ans. (c) — They are found free because they are **least** reactive.

C. Short Answer Questions (2–3 marks)

Q13. Differentiate between roasting and calcination. Give one chemical equation for each.

Ans. Roasting: sulphide ore is heated strongly in excess air; SO_2 is released.

$2ZnS + 3O_2 \rightarrow 2ZnO + 2SO_2$

Calcination: carbonate ore is heated strongly in limited air; CO_2 is released.

$ZnCO_3 \rightarrow ZnO + CO_2$

Q14. Write the chemical equations for the extraction of zinc from (i) zinc blende and (ii) calamine.

Ans. (i) Roasting: $2ZnS + 3O_2 \rightarrow 2ZnO + 2SO_2$; then reduction: $ZnO + C \rightarrow Zn + CO$

(ii) Calcination: $ZnCO_3 \rightarrow ZnO + CO_2$; then reduction: $ZnO + C \rightarrow Zn + CO$

Q15. How is mercury obtained from its ore? Write the balanced chemical equations.

Ans. Mercury is obtained from cinnabar (HgS). Being low in the activity series, the ore is simply heated in air. It first forms mercuric oxide, which on further heating gives mercury.

$2HgS + 3O_2 \rightarrow 2HgO + 2SO_2$; $2HgO \rightarrow 2Hg + O_2$

Q16. What is the thermit reaction? Write its equation and one use.

Ans. The reaction of iron(III) oxide with aluminium, which is so exothermic that iron is obtained in molten form.

$Fe_2O_3 + 2Al \rightarrow 2Fe + Al_2O_3 + Heat$

Use: joining railway tracks and repairing cracked machine parts.

Q17. Why can metals like Na, Mg and Al not be obtained by reduction with carbon? How are they obtained? Write the electrode reactions for sodium.

Ans. They have a greater affinity for oxygen than carbon, so carbon cannot remove oxygen from their oxides. They are obtained by electrolysis of their molten chlorides or oxides.

Cathode: $Na^+ + e^- \rightarrow Na$; Anode: $2Cl^- \rightarrow Cl_2 + 2e^-$

Q18. Silver articles become black and copper vessels turn green when left in air. Explain.

Ans. Silver reacts with sulphur compounds (H_2S) in air to form a black coating of silver sulphide. Copper reacts with moist carbon dioxide in air to form a green coating of basic copper carbonate.

Q19. What is an alloy? Name the constituents of solder and state the property that makes it suitable for welding electrical wires.

Ans. An alloy is a homogeneous mixture of two or more metals, or of a metal and a non-metal. Solder is made of lead and tin. Its **low melting point** makes it suitable for welding electrical wires.

Q20. Why is pure gold not used for making jewellery? What is meant by 22-carat gold?

Ans. Pure (24-carat) gold is very soft. It is alloyed with silver or copper to make it harder. 22-carat gold contains 22 parts of pure gold and 2 parts of copper or silver.

Q21. An ore on heating in air gives off a gas that turns acidified potassium dichromate green. What type of ore is it, and what process is being carried out? Name one such ore.

Ans. It is a **sulphide ore**, and the process is **roasting**. Example: zinc blende (ZnS) or cinnabar (HgS).

Q21a. Name the ore of lead. Write the equations for the extraction of lead from it.

Ans. Galena (PbS). Roasting: $2PbS + 3O_2 \rightarrow 2PbO + 2SO_2$; Reduction: $PbO + C \rightarrow Pb + CO$

Q21b. Name the metals obtained from (i) cassiterite, (ii) pyrolusite, (iii) bauxite, (iv) haematite. Which of these is extracted by electrolysis?

Ans. (i) Tin (ii) Manganese (iii) Aluminium (iv) Iron. **Aluminium** is extracted by electrolysis of alumina.

Q21c. Copper can be obtained from both copper glance and malachite. How do the first steps differ?

Ans. Copper glance (Cu_2S) is a sulphide ore — it is **roasted**, and copper forms by self-reduction: $2Cu_2O + Cu_2S \rightarrow 6Cu + SO_2$. Malachite is a carbonate ore — it is **calcined** to CuO, which is then reduced with carbon: $CuO + C \rightarrow Cu + CO$.

Q21d. Why is cryolite added during the extraction of aluminium from alumina?

Ans. Cryolite lowers the melting point of alumina and increases the electrical conductivity of the melt, saving energy.

D. Long Answer Questions (5 marks)

Q22. Explain the electrolytic refining of copper with the help of a labelled diagram. Write the reactions at the electrodes.

Ans. Draw the diagram from Section 7. Impure copper is made the **anode**, a thin strip of pure copper is the **cathode**, and **acidified copper sulphate** solution is the electrolyte. On passing current, copper from the anode dissolves as Cu^{2+} ions; an equal amount of pure copper deposits on the cathode. Soluble impurities go into solution; insoluble impurities collect below the anode as **anode mud**.

Anode: $Cu \rightarrow Cu^{2+} + 2e^-$; Cathode: $Cu^{2+} + 2e^- \rightarrow Cu$

Q23. Describe an activity to show that both air and water are necessary for the rusting of iron. Suggest two methods to prevent rusting.

Ans. Take three test tubes with clean iron nails. **A:** nails in tap water (air + water). **B:** nails in boiled distilled water with a layer of oil, corked (no air). **C:** nails with anhydrous $CaCl_2$, corked (no moisture). After a few days, only the nails in A rust. Hence both air and water are needed. (Draw the diagram from Section 8.)

Prevention: painting/oiling/greasing, galvanisation, chrome plating, alloying (stainless steel).

Q24. Draw a flowchart showing the steps involved in the extraction of metals of low, medium and high reactivity. Explain with one example each.

Ans. Draw the flowchart from Section 2. **Low:** HgS roasted in air → HgO → Hg on heating. **Medium:** ZnS roasted / $ZnCO_3$ calcined → ZnO, reduced with carbon → Zn. **High:** molten NaCl electrolysed → Na at cathode, Cl_2 at anode. All metals are then refined, usually by electrolytic refining.

Q25. (Case-based) Metal X is found as its sulphide ore and lies in the middle of the activity series. Its carbonate ore is also found in nature. (i) Identify X and name its two ores. (ii) Which process converts each ore into oxide? (iii) Write the equation for reduction of the oxide.

Ans. (i) X is **zinc**; ores: zinc blende (ZnS) and calamine ($ZnCO_3$). (ii) Zinc blende — roasting; calamine — calcination. (iii) $ZnO + C \rightarrow Zn + CO$